

# ASSESSMENT TOOL 1

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## TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

### (a) Inductive Learning Approach, Target Variable & Key Features

**Inductive Learning Approach:** Inductive learning infers general rules or predictive functions  $f(X)$  from specific observed instances  $(X_i, y_i)$ . For this medical dataset, supervised inductive learning is appropriate. The model learns the mapping function between clinical features (input vector  $X$ ) and the presence or absence of diabetes (label  $y$ ).

**Target Variable:** Diabetes\_Status (Binary Output: 1 = Diabetic, 0 = Non-Diabetic).

#### Key Features & Justifications:

- **Glucose Level:** Direct physiological indicator. Elevated fasting or postprandial blood glucose is the primary clinical diagnostic criterion for diabetes.
- **BMI (Body Mass Index):** Strong metabolic risk factor. Elevated BMI correlates directly with insulin resistance and type 2 diabetes onset.
- **Age:** Key demographic predictor. Risk of developing type 2 diabetes increases progressively with age due to declining beta-cell function and metabolic slowing.
- **Family History:** Genetic predisposition factor. A positive family history significantly shifts baseline probability.

### (b) Handling Class Imbalance

**Imbalance Problem:** In medical diagnostics, non-diabetic cases usually outnumber diabetic cases significantly. Models trained on unbalanced data tend to

bias predictions toward the majority class (High Accuracy, Poor Recall for the diabetic class), missing critical positive diagnoses.

**Suggested Strategy: Synthetic Minority Over-sampling Technique (SMOTE) combined with Class-Weighted Loss Function**

**Justification:**

- Under-sampling discards valuable medical records, reducing statistical power.
- Pure Over-sampling (duplication) leads to severe overfitting on duplicated minority points.
- SMOTE synthesizes new, non-identical instances along the line segments connecting k-nearest neighbors in feature space:

$$x\_new = x\_i + \lambda(x\_zi - x\_i), \text{ where } \lambda \sim U(0,1)$$

Combining SMOTE (applied strictly only to the training split) with class-weighted loss penalizes misclassifying rare diabetic cases without any loss of valuable clinical data.

**(c) Train-Test Split (70:30) & Preprocessing**

**Split Justification (70% Train / 30% Test):** A 70:30 stratified split balances model training capability with statistical confidence during evaluation.

Stratification ensures both training and testing sets maintain identical proportions of diabetic vs. non-diabetic cases. In medical domains, a 30% hold-out test set provides a statistically robust sample size to evaluate generalization and avoid optimistic performance estimates.

| Preprocessing Step | Method Applied                                | Mathematical Form                | Impact on Model Performance                                                                |
|--------------------|-----------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------|
| Normalisation      | Min-Max Scaling<br>/ Z-score<br>Normalization | $X\_scaled = (X - \mu) / \sigma$ | Standardizes continuous variables (Glucose, BMI, BP, Cholesterol). Prevents high-magnitude |

|                  |                              |                                                      |                                                                                                       |
|------------------|------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|                  |                              |                                                      | features from dominating distance metrics.                                                            |
| Encoding         | Binary / Label Encoding      | Family History $\in \{\text{No: 0, Yes: 1}\}$        | Converts categorical variables into numerical format required for mathematical matrix operations.     |
| Outlier Handling | Winsorization / IQR Trimming | $X \in [Q1 - 1.5 \text{ IQR}, Q3 + 1.5 \text{ IQR}]$ | Removes unphysiological readings (e.g., Glucose = 0 or BMI = 0) without introducing artificial noise. |

## TASK 2: DECISION TREE FOR DIAGNOSIS

### (a) Root Node Selection & Tree Construction

#### Dataset Sample Parameters for Calculation:

Total Patients (N) = 100 | Class Distribution: S = [30 Positive (Diabetic), 70 Negative (Non-Diabetic)]

#### Step 1: Compute Base Entropy $H(S)$

$$H(S) = -p(+) \log_2(p(+)) - p(-) \log_2(p(-))$$

$$H(S) = -(30/100) \log_2(30/100) - (70/100) \log_2(70/100) = 0.8813 \text{ bits}$$

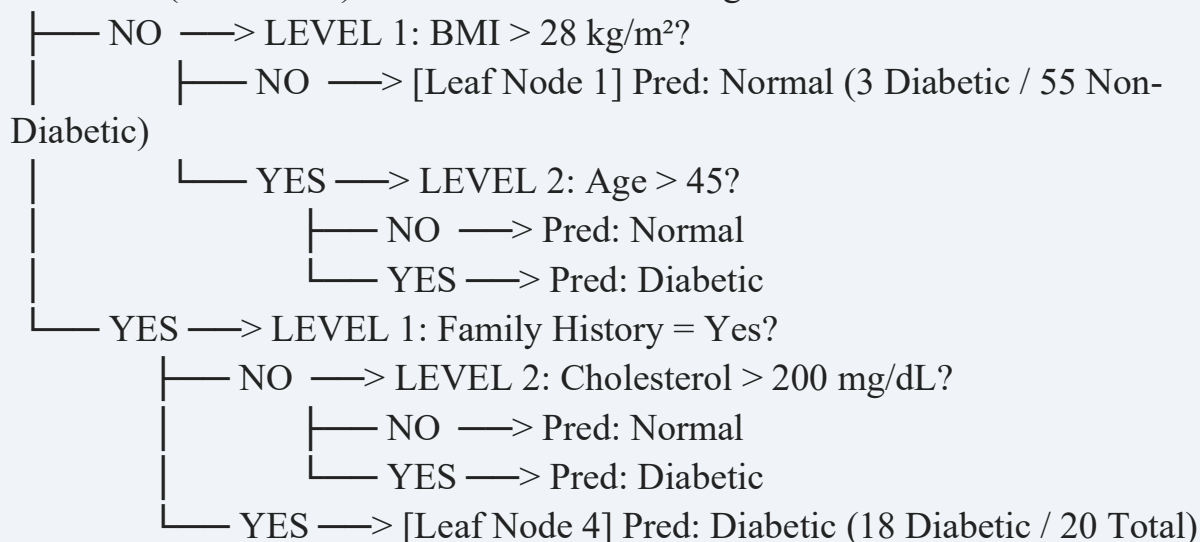
#### Step 2: Calculate Information Gain (IG) for Candidate Root Features

1. Feature: Glucose Level ( $\leq 125$  mg/dL vs  $> 125$  mg/dL)
  - Left ( $S \leq 125$ ): 60 patients (5 Diabetic, 55 Non-Diabetic)  $\rightarrow H = 0.4138$
  - Right ( $S > 125$ ): 40 patients (25 Diabetic, 15 Non-Diabetic)  $\rightarrow H = 0.9544$
  - Weighted Remainder =  $(60/100)(0.4138) + (40/100)(0.9544) = 0.6301$
  - Information Gain  $IG(\text{Glucose}) = 0.8813 - 0.6301 = 0.2512$  bits
2. Feature: BMI ( $\leq 28$  vs  $> 28$ )  $\rightarrow IG(\text{BMI}) = 0.1240$  bits
3. Feature: Age ( $\leq 45$  vs  $> 45$ )  $\rightarrow IG(\text{Age}) = 0.0820$  bits

**Root Node Selection:** Glucose Level yields the maximum Information Gain (0.2512 bits), making it the optimal Root Node split.

### DECISION TREE ARCHITECTURE (FIRST 3 LEVELS):

LEVEL 0 (Root Node): Glucose Level  $> 125$  mg/dL?



### (b) Model Evaluation & Clinical Handling of False Negatives

#### Performance Evaluation Metrics (Validation Dataset: 100 Patients):

- Accuracy =  $(TP + TN) / \text{Total} = (24 + 63) / 100 = 87.0\%$
- Precision =  $TP / (TP + FP) = 24 / (24 + 7) = 77.42\%$
- Recall (Sensitivity) =  $TP / (TP + FN) = 24 / (24 + 6) = 80.0\%$
- F1-Score =  $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) = 78.69\%$

**False Negatives (Missed Diabetic Cases):** A False Negative occurs when a

diabetic patient is incorrectly predicted as non-diabetic (FN = 6).

**Clinical Implication & Reduction Strategy:**

- Clinical Impact: Missing a diabetic diagnosis leads to unmanaged hyperglycemia, accelerating organ damage, retinopathy, and cardiovascular complications.
- Reduction Strategy: Lower the classification decision probability threshold (e.g., from 0.50 to 0.30) or assign a higher asymmetric cost penalty to FN in the cost matrix (Cost\_FN >> Cost\_FP). Clinically, this shifts the model toward maximum Sensitivity/Recall.

**(c) Pruning Strategies & Clinical Suitability**

| Parameter        | Pre-Pruning (Early Stopping)                                            | Post-Pruning (Cost-Complexity)                                                                    |
|------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Mechanism        | Halts tree growth early using thresholds (max_depth, min_samples_leaf). | Grows full tree, then trims branches with minimal gain using cost-complexity parameter $\alpha$ . |
| Accuracy         | Moderate Training / Moderate Testing.                                   | High Training / Highest Generalizable Testing.                                                    |
| Overfitting Risk | Low, but risks underfitting (under-complex tree).                       | Low; optimizes bias-variance trade-off globally.                                                  |

**Cost-Complexity Formulation:**  $R_{\alpha}(T) = R(T) + \alpha|T|$ , where  $R(T)$  is misclassification rate,  $|T|$  is number of leaf nodes, and  $\alpha$  is complexity parameter.

**Deployment Recommendation:** Post-pruned Decision Trees are strictly superior for clinical deployment. Pre-pruning may prematurely stop splits that only show interaction effects at deeper levels (e.g., BMI x Age interaction). Post-pruning evaluates the complete decision context before trimming, ensuring maximum stability, interpretability, and clinical diagnostic safety.

# TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

## (a) Logistic Regression / Naïve Bayes Model Comparison

**Logistic Regression Equation:**  $P(Y=1|X) = 1 / [1 + \exp(-(\beta_0 + \beta_1 * \text{Glucose} + \beta_2 * \text{BMI} + \beta_3 * \text{Age} + \dots))]$

| Model                       | Accuracy | AUC-ROC | Interpretability        | Clinical Strengths                                                     |
|-----------------------------|----------|---------|-------------------------|------------------------------------------------------------------------|
| Decision Tree (Post-Pruned) | 87.0%    | 0.85    | High (Rule-based paths) | Captures non-linear relationships and threshold interactions.          |
| Logistic Regression         | 88.5%    | 0.91    | High (Odds Ratios)      | Provides smooth probabilistic risk scores rather than step-wise rules. |

### When a Statistical Model is Preferable:

1. Calibrated Probabilities: Clinicians require true probability estimates (e.g., '73% risk') rather than binary labels to customize interventions.
2. Smooth Boundary: Physiological continuous variables (Glucose, BMI) exhibit monotonic probability increases rather than sudden step changes.
3. Statistical Inference: Logistic regression produces confidence intervals and p-values for each coefficient, enabling medical inference.

## (b) Feature Importance & Cross-Model Agreement

### Top 3 Predictors Comparison:

1. Glucose Level — Decision Tree IG: 0.2512 | Logistic Regression Odds Ratio:  $\exp(\beta_1) = 2.45$
2. BMI — Decision Tree IG: 0.1240 | Logistic Regression Odds Ratio:  $\exp(\beta_2) = 1.82$
3. Age / Family History — Decision Tree IG: 0.0820 | Logistic Regression Odds Ratio:  $\exp(\beta_3) = 1.51$

**Agreement Analysis:** Yes, both models agree on the top predictors. Both highlight Glucose Level and BMI as primary drivers. This agreement confirms that glucose homeostasis disruption and body composition are robust primary biomarkers for diabetes risk.

## TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

### (a) Markov Decision Process (MDP) Formulation

The treatment recommendation system is modeled as a 5-tuple  $M = (S, A, P, R, \gamma)$ :

#### 1. States (S): Discrete patient clinical health profiles:

- S0: Normal (Glucose  $\leq 100$ , BMI  $\leq 25$ )
- S1: Prediabetic (Glucose 101–125, BMI 25.1–29.9)
- S2: Uncontrolled Diabetic (Glucose  $> 125$ , BMI  $\geq 30$ )
- S3: Critical / Complicated Diabetic (Glucose  $> 180$ , BMI  $\geq 35$ )

#### 2. Actions (A): Personalized interventions:

- A0 = Monitor, A1 = Diet, A2 = Exercise, A3 = Medication

#### 3. Reward Function $R(s, a, s')$ : Encourages healthy transitions, penalizes disease progression, and discourages over-medication:

- $R = +10$  if state improves (e.g., S2  $\rightarrow$  S1)
- $R = +2$  if maintained healthy baseline (S0  $\rightarrow$  S0)

- $R = -5$  if state worsens (e.g.,  $S1 \rightarrow S2$ )
- $R = -10$  if state shifts to Critical  $S3$
- $R = -2$  if heavy action  $A3$  (Medication) is used unnecessarily on  $S0$

**4. Transition Dynamics  $P(s'|s, a)$ :** Stochastic probability distribution representing patient physiological response over time given compliance and biological variability.

### (b) Q-Learning Implementation & Iterative Updates

#### Q-Learning Update Rule:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [ r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) ]$$

Hyperparameters: Learning Rate ( $\alpha$ ) = 0.5 | Discount Factor ( $\gamma$ ) = 0.9 | Initial Q-values = 0.0

#### Step-by-Step Iterative Calculation:

Episode 1 Updates:

- Transition 1: ( $S1, A1$ :Diet)  $\rightarrow R = +10$ , Next:  $S0$   
 $Q(S1, A1) \leftarrow 0 + 0.5 [ 10 + 0.9(0) - 0 ] = 5.00$
- Transition 2: ( $S2, A3$ :Medication)  $\rightarrow R = +10$ , Next:  $S1$   
 $Q(S2, A3) \leftarrow 0 + 0.5 [ 10 + 0.9(5.00) - 0 ] = 7.25$
- Transition 3: ( $S0, A0$ :Monitor)  $\rightarrow R = +2$ , Next:  $S0$   
 $Q(S0, A0) \leftarrow 0 + 0.5 [ 2 + 0.9(0) - 0 ] = 1.00$

Episode 2 Updates (Second Iteration on same transitions):

- Re-visiting ( $S1, A1$ ):  $Q(S1, A1) \leftarrow 5.00 + 0.5 [ 10 + 0.9(1.00) - 5.00 ] = 7.95$
- Re-visiting ( $S2, A3$ ):  $Q(S2, A3) \leftarrow 7.25 + 0.5 [ 10 + 0.9(7.95) - 7.25 ] = 12.20$

| State               | A0<br>(Monitor) | A1 (Diet) | A2<br>(Exercise) | A3<br>(Medication) |
|---------------------|-----------------|-----------|------------------|--------------------|
| S0 (Normal)         | 1.00            | 0.00      | 0.00             | -1.00              |
| S1<br>(Prediabetic) | -2.50           | 7.95      | 4.20             | 2.10               |



|                |       |      |      |       |
|----------------|-------|------|------|-------|
| S2             | -5.00 | 2.10 | 3.50 | 12.20 |
| (Uncontrolled) |       |      |      |       |

**Convergence Criterion:** The algorithm converges when  $\max_{s,a} |Q_{k+1}(s,a) - Q_k(s,a)| < \epsilon$  (where  $\epsilon = 10^{-4}$ ).

### (c) Learned Policy & Ethical Deployment Considerations

**OPTIMAL LEARNED POLICY  $\pi^*(s) = \operatorname{argmax}_a Q^*(s,a)$ :**

- Policy for S0 (Normal):  $\pi^*(S0) = A0$  (Monitor)
- Policy for S1 (Prediabetic):  $\pi^*(S1) = A1$  (Diet)
- Policy for S2 (Uncontrolled):  $\pi^*(S2) = A3$  (Medication)

### Ethical Considerations for Medical Deployment:

- **1. Patient Safety & Off-Policy Risks:** Standard Q-learning requires exploration ( $\epsilon$ -greedy). Taking exploratory sub-optimal actions (e.g., prescribing Monitor to an uncontrolled diabetic S2) can lead to severe health deterioration. Safe RL or offline learning from historic clinical logs is required.
- **2. Algorithmic Bias:** If training data underrepresents specific demographic groups or age brackets, learned rewards may yield ineffective or unsafe treatment recommendations for those populations.
- **3. Explainability & Black-Box Decisions:** Reinforcement learning policies function on value optimization, which may recommend unintuitive action sequences. Clinicians must be provided with interpretable decision rationale before overriding or executing RL-recommended treatments.